

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates the correct sample mean	AO1.1b	B1	$\bar{x} = \frac{3540}{100}$ $= 35.4$
	Finds correct z value to at least 3 significant figures. Condone -2.326... Can be implied by a correct confidence interval.	AO1.1a	M1	$z = 2.32634757$ $\bar{x} \pm z \sqrt{\frac{\sigma^2}{n}}$
	Uses formula for confidence interval with $\sqrt{\frac{10}{100}}$	AO1.1a	M1	$= 35.4 \pm 2.326 \sqrt{\frac{10}{100}}$ $= (34.66, 36.14) \text{ or } 35.4 \pm 0.7357$
	Obtains confidence interval CAO	AO1.1b	A1	
(b)	States null hypothesis is rejected as the confidence interval does not contain 38. OE Follow through their confidence interval.	AO3.5a	E1F	Dante rejects the null hypothesis because 38 is outside the confidence interval.
				Total 5 marks

Q2.

(a) $L \sim N(1005, 15^2)$

$V(\text{pack}) = 15^2 / 12 \text{ or } 225 / 12 \text{ or } 75 / 4$
CAO

or

18.7 to 18.8

AWFW (18.75)

OR

$SD(\text{pack}) = 15 / \sqrt{12} \text{ or } 15 / 2\sqrt{3} \text{ or } 5\sqrt{3} / 2$
CAO; OE

or

4.3 to 4.4

AWFW (4.33013)

B1

$$P(L < 1000) = P\left(\frac{1000 - 1005}{15/\sqrt{12}}\right) =$$

Standardising 1000 using 1005 and 15 / $\sqrt{12}$ OE; allow (1005 - 1000)

M1

$$P(Z < -1.1547) = 1 - P(Z < 1.1547) =$$

Correct area change
May be implied by a correct answer
or an answer < **0.5**

m1

$$1 - (0.87698 \text{ to } 0.87493) = \underline{\mathbf{0.123 \text{ to } 0.126}}$$

AWFW (0.12411)
(1 - answer) $\Rightarrow$ B1 M1 max

A1

4

(b) (i) 99% (0.99) $\Rightarrow z = \underline{\mathbf{2.57 \text{ to } 2.58}}$

AWFW (2.5758)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

Used with z (2.05 to 2.58),
 $\bar{x}$ (4.65) & σ (0.15)
and $\div \sqrt{n}$ with $n > 1$

M1

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Thus $4.65 \pm 2.5758 \times \frac{0.15}{\sqrt{24}}$

z (2.05 to 2.06 or 2.32 to 2.33
or 2.57 to 2.58),
 $\bar{x}$ (4.65) & σ (0.15)
and $\div \sqrt{24 \text{ or } 23 \text{ or } 12 \text{ or } 11}$

A1

Hence $\underline{\mathbf{4.65 \pm 0.08}}$

CAO / AWRT

OR

$\underline{\mathbf{(4.57, 4.73)}}$

AWRT

A1

4

- (ii) **Clear correct comparison of 4.5 with LCL or CI**
 (eg $4.5 < \text{LCL}$ or its value
 or $4.5 < \text{CI}$ or its limits

*F on CI only providing $\text{LCL} > 4.5$
 (ie whole of CI > 4.5)
 Quoting values for LCL or for CI
 is **not** required
 BF0 for '4.5 is outside CI'; OE*

BF1

so

Agree with manufacturer's specification
 OE; dependent on previous BF1

Bdep1

2

[10]

Q3.

- (a) 98% (0.98) $\Rightarrow z = \underline{2.32 \text{ to } 2.33}$
 AFWW (2.3263)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$
*Used with z (2.05 to 2.58),
 $\bar{x}$ (19.9) and σ (0.4)
 and $\div \sqrt{n}$ with $n > 1$*

M1

Thus $19.9 \pm 2.3263 \times \frac{0.4}{\sqrt{25}}$
 z (2.05 to 2.06 or 2.32 to 2.33 or 2.57 to 2.58),
 $\bar{x}$ (19.9) and σ (0.4) and $\div \sqrt{25 \text{ or } 24}$

A1

Hence $\underline{19.9 \pm 0.2}$
 CAO / AWRT (0.186104)

or

$\underline{(19.7, 20.1)}$
 AWRT

A1

4

- (b) **Clear correct comparison of 20 with CI**



F on CI providing it contains 20

eg 20 is within CI **or** $LCL < 20 < UCL$

*Quoting values for CI is **not** required*

BF1

so

Agree with claim **or** no reason to doubt claim

OE; dependent on previous BF1

Bdep1

2

[6]

Q4.

(a) (i) $\bar{x} = \frac{497.5}{25} = \underline{19.9}$

CAO

B1

98% (0.98) $\Rightarrow z = \underline{2.32 \text{ to } 2.33}$
AWFW (2.3263)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

*Used with z (2.05 to 2.58),
 $\bar{x}$ (497.5 or 19 to 21) and σ (0.4)
and $\div \sqrt{n}$ with $n > 1$*

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M1

Thus $19.9 \pm 2.3263 \times \frac{0.4}{\sqrt{25}}$
 z (2.05 to 2.06 or 2.32 to 2.33 or 2.57 to 2.58),
 $\bar{x}$ (19.9) and σ (0.4) and $\div \sqrt{25 \text{ or } 24}$

A1

Hence $\underline{19.9 \pm 0.2}$
CAO / AWRT (0.186104)

or

$\underline{(19.7, 20.1)}$
AWRT

A1

- (ii) **Clear correct comparison of 20 with CI**
F on CI providing it contains 20

eg 20 is within CI **or** $LCL < 20 < UCL$

*Quoting values for CI is **not** required*

BF1

Agree with claim **or** no reason to doubt claim

OE; dependent on previous BF1

Bdep1

2

- (b) (i) $Y \sim N(25.25, 0.35^2)$

Accept percentage equivalent probabilities

$$V(\text{mean}) = 0.35^2 / 10 \text{ or } 0.0122 \text{ to } 0.0123$$

CAO / AFWW (0.01225)

or

$$SD(\text{mean}) = 0.35 / \sqrt{10} \text{ or } 0.11 \text{ to } 0.111$$

CAO / AFWW (0.11068)

B1

$$P(\bar{Y} < 25) = P\left(Z < \frac{25 - 25.25}{0.35/\sqrt{10}}\right)$$

*Standardising 25 using 25.25 and $0.35 / \sqrt{10}$ OE
but allow $(25.25 - 25)$*

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$$= P(Z < -2.25877) = 1 - P(Z < 2.25877)$$

Correct area change

*May be implied by a **correct** answer **or** an answer < 0.5*

m1

$$= 1 - (0.98809 \text{ to } 0.98778)$$

$$= \underline{0.011 \text{ to } 0.013}$$

AWFW (0.01195)

$(0.987 \text{ to } 0.989) \Rightarrow B1 M1 m0 A0$

A1

4

(ii)
$$P(Y > 25) = P\left(Z > \frac{25 - 25.25}{0.35}\right)$$

Standardising 25 using 25.25 and 0.35 but allow (25.25 – 25)

M1

$$= P(Z > -0.71429) = P(Z < 0.71429)$$

$$= \underline{\mathbf{0.761 \text{ to } 0.764}}$$

AWFW (0.76247)
(0.236 to 0.239) $\Rightarrow$ M1 A0

A1

$$P(Y > 25 \text{ in each of } 10) = p^{10}$$

Any p^{10} providing $0 < p < 1$

May be implied by a **correct** answer

M1

$$= \underline{\mathbf{0.065 \text{ to } 0.068}}$$

AWFW (0.06641)

A1

4

[15]

Q5.

(a) $\bar{x} - ns = (45.8 - n \times 24.0) < 0$

SC: Accept quoted values of (-4 to -1) ($n = 2$) or
(-28.5 to -23.5) ($n = 3$) (both AFWW)

and

Allow (45 to 47) and any multiple of (23.5 to 24.5) which
gives value < 0

Must clearly state the value of a numerical expression

M1

negative salaries are impossible

OE; must be in context

Negative values impossible $\Rightarrow$ A0

A1

2

Alternative solution 1

$$P(X < 0 \mid N(45.8, 24.0^2)) = P(Z < -1.91)$$

Standardising 0 using 45.8 & 24.0

(M1)

$$= 0.027 \text{ to } 0.03$$

In addition to probability within range, must state that



(A1)

(2)

Alternative solution 2

$$P(X > 60 \mid N(45.8, 24.0^2)) = P(Z > 0.59)$$

Standardising 60 using 45.8 & 24.0

(M1)

$$= 0.27 \text{ to } 0.28$$

In addition to probability within range, must compare
calculated value to $6/50 = 0.12$ OE

(A1)

(2)

- (b) (i) Large sample or $n > 25$ or 30 or $n = 50$
so

OE

B1

CLT applies

Must indicate CLT; dependent on B1
Indication that other than sample mean is normally
distributed $\Rightarrow$ Bdep0

Bdep1

2

- (ii) 99% (0.99) $\Rightarrow z = 2.57$ to 2.58
AWFW (2.5758)

B1

CI for μ is $\bar{x} \pm z \times \frac{s}{\sqrt{n}}$
Used with (45.8 & 24.0) and
 $z(1.64 \text{ to } 2.58) \text{ \& } \div \sqrt{n}$ with $n > 1$

M1

Thus $45.8 \pm 2.5758 \times \frac{24.0}{\sqrt{50}}$
F on $z(1.64 \text{ to } 1.65 \text{ or } 2.32 \text{ to } 2.33$
or $2.57 \text{ to } 2.58)$ and $\div \sqrt{50 \text{ or } 49}$

AF1

Hence $45.8 \pm (8.7 \text{ to } 8.8)$ or $45800 \pm (8700 \text{ to } 8800)$
OR

(37.0) to 37.1, 54.5 to 54.6) or (37000 to 37100, 54500 to 54600)

CAO / AFWW (8.74)

Ignore (absence of) quoted units

AWFW

A1

4

- (c) Clear correct comparison of 55 or 55000 with c's UCL or CI

*Accept 55000 compared with c's 54.5 to 54.6
(ie different units)*

B1

(6/50 or 0.12 or 12%) $\neq$ 0.25 or 25%

OE; correct comparison mentioning both 12% and 25%

B1

Reject both / each of the two claims

Dependent on B1 B1

Bdep1

3

Additional comment illustrations

It / (claimed) mean / (claimed) value $>$ UCL / CI

Must indicate 55 or 55000

(B0)

99% have (mean) weights between CLs so...

(B0)

Any comparison of 60 (£60 000) with UCL / CI

Value of 60 does not refer to mean

(B0)

$P(\bar{X} > 60 \mid N(45.8, 24.0^2)) = P(Z < 0.59)$

$= (0.27 \text{ to } 0.28) > 6/50 = 0.12$

Assumes salaries $\sim N$; cf (a)(ii)

(B0)

[11]

Q6.

- (a) 98% (0.98) $\Rightarrow z = \underline{2.32 \text{ to } 2.33}$

AWFW

(2.3263)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

Used with z (2.05 to 2.58), $\bar{x}$ (5.05, 5050 or 181.8),
 σ (0.0075, 0.075, 0.75, 7.5 or 75) and $\div \sqrt{n}$ with $n > 1$

M1

Thus $5.05 \pm 2.3263 \times \frac{0.075}{\sqrt{36}}$
 z (2.05 to 2.06 or 2.32 to 2.33 or 2.57 to 2.58),
 $\bar{x}$ (5.05) & σ (0.075) or x (5050) & σ (75)
 and $\div \sqrt{36}$ or 35

A1

Hence **5.05 $\pm$ 0.03 or 5050 $\pm$ 30**

OR

(5.02, 5.08) or (5020, 5080)

CAO / AWRT

Dependent on previous A1 so can be
 scored with $z \neq 2.32$ to 2.33

Ignore (absence of) quoted units

AWRT to 3sf accuracy

Adep 1

4

Note Use of t (2.43 to 2.72) $\Rightarrow$ B1 B0 M1 A0 A0 max

(b) **Clear correct comparison of 5 or 5000 with LCL or CI**

so

agree with (first) claim (about mean)

Dependent on Adep1 in (a)

Must use consistent units

Adep 1

(8/36 or 0.22 or 22%) v (1/10 or 0.10 or 10%)

or 8 v 3.6 (3 to 4)

Mention of a value on LHS **and** a value on RHS

B1

so

8/36 OE > / $\neq$ 1/10 OE

so

disagree with (second) claim (about individuals)

Dependent on B1

Explicit comparison of values and correct conclusion

Bdep 1

3

Notes 1 It / (claimed) mean / (claimed) value < LCL / CI $\Rightarrow$ Adep0
Must indicate 5 or 5000

2 98% have (mean) weights between CLs so ... $\Rightarrow$ Adep0

3 Any reference to CI for second claim $\Rightarrow$ B0 Bdep0 Claim refers to individual bottles

[7]

Q7.

(a)

$$\left. \begin{array}{l} Y \sim N(\mu, 4) \\ n = 16, \bar{y} = 450 \end{array} \right\}$$

(known variance) use $\Rightarrow$ z

For 95% CI $z_{crit} = 1.96$

$$\left. \begin{array}{l} 450 \pm 1.96 \times \frac{2}{\sqrt{16}} \\ 450 \pm 0.98 \\ 450 \pm 0.98 \end{array} \right\}$$

B1

M1

(449, 451)

awrt

A1

3

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(b) (i) $X \sim N(\mu, \sigma^2)$

(unknown variance) $\Rightarrow$ use t_{n-1}

$$\left. \begin{array}{l} n = 9 \quad \& \quad \bar{x} = \frac{4950}{9} = 550 \\ s^2_{n-1} = \frac{334}{8} = 41.75 \quad (s_{n-1} = 6.461) \end{array} \right\}$$

both

B1

For 90% CI $t_{crit} = 1.860$

B1

$$550 \pm 1.860 \times \frac{\sqrt{41.75}}{\sqrt{9}}$$

$$\left\{ \begin{array}{l} \text{their } \bar{x} \pm t_8 \times \frac{\text{their } s_{n-1}}{\sqrt{9}} \\ \text{(must have at } t_8 - \text{value)} \end{array} \right.$$

M1

$$550 \pm 40$$

A1 ft

(546,554)

awrt

A1

5

(ii) 545 not in 90% CI

∴ Reject claim

Evidence to suggest that **mean** content of a bottle of tomato sauce is not 545 grams and hence to reject Holly's claim

Alternative

$$H_0 : \mu_X = 545$$

$$H_1 : \mu_X \neq 545$$

$$t = \frac{550 - 545}{\sqrt{41.75 \times \frac{75}{9}}} = 2.32$$

B1ft

(dep on (b)(i))

at 10% level of significance.

$$t_{crit} = 1.86 < 2.32$$

Reject H_0

B1

Comment in context E1

E1 ft

(dep on (b)(i))

10% level of significance.

B1

3

Alternatives (such as):

Claim justified at 1% level of significance

$$t_{crit} = 3.355 > 2.32$$

Accept H_0

1% significance level

[11]

Q8.

- (a) (i) Area / $F(x) = 10u \times 0.01\pi$ (OE)
Shown by any correct method

B1

$$= 1 \Rightarrow u = \frac{10}{\pi}$$

Alternatives: $f = \frac{1}{10u}$ B1

Bdep1

or

$$u = \frac{10}{\pi} \Rightarrow F(x) = 1$$

Show $u = \frac{10}{\pi}$ or show $\frac{1}{10u} = 0.01\pi$ Bdep1

(Bdep1)

2

(ii) $E(X) = \frac{1}{2}(11u + u) = 6u = 6 \times \frac{10}{\pi} = \frac{60}{\pi}$

Must be in terms of π (eg $60\pi^{-1}$)

B1

1

$$\text{Var}(X) = \frac{1}{12}(b-a)^2$$

$$\text{Var}(X) = \frac{1}{12}(11u - u)^2$$

$$= \frac{1}{12} \times 100 \times \frac{100}{\pi^2} = \frac{100^2}{12\pi^2}$$

Alternatives:

$$\frac{10000}{12\pi^2} = \frac{5000}{6\pi^2} = \frac{2500}{3\pi^2} = \left(\frac{50}{\pi\sqrt{3}}\right)^2 = \frac{(\text{AWRT } 833)}{\pi^2}$$

Must be in terms of π

B1

1

(iii) $C = \pi \left(X + \frac{10}{\pi} \right)$



$$E(C) = \left. \begin{aligned} &\pi \times [\text{their } E(X)] + 10 \\ &\pi \times \frac{60}{\pi} + 10 \end{aligned} \right\}$$

Their **numerical** value of $E(X)$ used correctly
Must have a multiplier of π or 2π

M1

$$= 70$$

CAO

A1

$$\text{Var}(C) = \pi^2 \times \frac{100^2}{12\pi^2} = \frac{100^2}{12}$$

$$\pi^2 \times [\text{their } \text{Var}(X) > 0]$$

Must have a multiplier of π^2 or $4\pi^2$

M1

$$= 833\frac{1}{3} (833.\dot{3})$$

Alternatives:

$$\frac{10000}{12} = \frac{5000}{6} = \frac{2500}{3}$$

Must be exact: 833.3 gets A0

A1

4

(b) $n = 100$ and $\bar{a} = 40.5$

$$\mu = \left. \begin{aligned} &40.5 \pm z \times \frac{\sqrt{25}}{\sqrt{100}} \\ &40.5 \pm 1.0 \end{aligned} \right\}$$

95% CI for

For $z = 1.96$

B1

$z = 1.96$ or 1.64 to 1.65 only

M1

$$= (39.5, 41.5)$$

AWRT

A1

3

[11]

Q9.

$$95\% \Rightarrow z = 1.96$$

B1

Require $2 \times \frac{1.96\sigma}{\sqrt{n}} \leq 0.2\mu$

Used; may be implied

Allow 'no 2 ×'

Allow '= sign' throughout

M1

Thus $2 \times \frac{1.96}{\sqrt{n}} \times \frac{\mu}{2} \leq 0.2\mu$

Use of $\sigma = \frac{\mu}{2}$; may be implied

Allow 'no 2 ×'

M1

Thus $\sqrt{n} \geq \frac{1.96}{2}$

Attempt at solution for $\sqrt{n}$ or n

M1

Thus $n \geq 96.04$

Thus, to nearest 10; $n = 100$

CAO

A1

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[5]

Q10.

(a) Represent a random sample

Or equivalent

B1

Mean = $\bar{x} = \frac{3155}{50} = 63.1$

CAO

B1

Variance = $s^2 = \frac{7180.5}{49} = 146.5 \text{ to } 146.6$

AWFW $\left(\frac{7180.5}{50} = 143.61 \right)$

B1

3

- (b) 99% (0.99) $\Rightarrow z = 2.57$ to 2.58
AWFW (2.5758)
 $t_{49} (0.995) = 2.67$ to 2.68

B1

CI for μ is $\bar{x} \pm z/t \times \frac{s}{\sqrt{n}}$
Used
Must have $\sqrt{n}$ with $n > 1$

M1

Thus 63.1 $\pm 2.5758 \times \frac{\sqrt{146.5}}{\sqrt{50}}$
F on $\bar{x}$, s^2 and z/t

A1F

Hence 63.1 $\pm$ (4.39 to 4.42)
or (58.6 to 58.8, 67.4 to 67.6)
CAO & AFWW
Dependant on A1F with correct s^2 / s
AWFW (58.7, 67.5)

A1dep

Note:

Use of t gives 63.1 $\pm$ (4.57 to 4.59)
or (58.5 to 58.6, 67.6 to 67.7)

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- (c) 60 minutes is within C1
Or equivalent; F on (b)
Any reference to 63.1 $\Rightarrow$ B0F

B1F

Wyn's belief doubtful/incorrect

or

Wyn's belief may be correct

Or equivalent

Dependent on B1F

B1 dep

2

- (d) 1/100 or 0.01 or 1%
CAO; not 0.01%

B1

1

[10]

Q11.

(a) $\text{Mean} = \frac{1620}{30} = 54$

CAO; cannot be gained in (b)

B1

1

(b) 98% (0.98) $\Rightarrow z = 2.32$ to 2.33

AWFW (2.3263)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$
Used
Must have $\sqrt{n}$ with $n > 1$

M1

Thus $54 \pm 2.3263 \times \frac{8}{\sqrt{30}}$
F on $\bar{x}$ (but **not** 1620) and z only
Allow $\bar{x} = 54$ even if B0 in (a)

A1F

Hence $54 \pm (3.38 \text{ to } 3.42)$
CAO & AWFW (54 & 3.4)

or

(50.58 to 50.62, 57.38 to 57.42)

AWFW (50.6, 57.4)

A1

Notes:

Use of $n = 1$ in (b) must not be deemed as answer to (c)

Use of $n = 1$ in (b) followed by use of $n = 1$ in (c)

$\Rightarrow$ (b) B1, (c) M1 A1 max

Use of $n = 1$ with (b) or (c) not identified

$\Rightarrow$ (b) B1, (c) 0 max

4

- (c) Repeat of structure in (b) but with $n = 1$
and $1.96 \leq z \leq 3.03$

Or equivalent
CAO & AFWW (54 & 18.6)

M1

Thus $54 \pm (18.56 \text{ to } 18.64)$

If z-value incorrect, then must use
 $54 \pm 8 \times [z \text{ from (b)}]$

A1F

or

(35.36 to 35.44, 72.56 to 72.64)
AWFW (35.4, 72.6)

Note:
Accept sensible non-symmetric intervals such as:
(0, $54 + 2.0537 \times 8$)
= (0, 70.4 to 70.5)

2

[7]

Q12.

(a) 98% (0.98) $\Rightarrow z = 2.32 \text{ to } 2.33$
AWFW (2.3263)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$
Used
Must have $\sqrt{n}$ with $n > 1$

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M1

Thus $1030 \pm 2.3263 \times \frac{50}{\sqrt{20}}$
F on z only

A1F

Hence 1030 ± 26
CAO & AWRT

or (1004, 1056)
AWRT

A1

4

(b) Whole of confidence interval is **above 1000**

F on (a)
Or equivalent

B1F

so

Agree with claim

F on (a)
Or equivalent
Dependent on previous B1F

B1Fdep

2

[6]

Q13.

(a) Mean $\frac{470}{10} = 47$
CAO

B1

98% (0.98) $\Rightarrow z = 2.32$ to 2.33
AWFW (2.3263)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$
Used
Must have $\sqrt{n}$ with $n > 1$

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M1

Thus $47 \pm 2.3263 \times \frac{15}{\sqrt{10}}$
F on $\bar{x}$ and z only

A1F

Hence 47 ± 11.0 to 11.1
CAO & AWRT (accept 11)

Or
(35.9 to 36.0, 58.0 to 58.1)
AWRT (accept 36 & 58)

A1

5

(b) $Y \sim N(108, 28^2)$

Variance of $\bar{Y}_{40} = 28^2/40 = 19.6$

CAO

Stated or used

B1

$\sqrt{\text{SD of } \bar{Y}_{40}} = 28/\sqrt{40} = 4.425 \text{ to } 4.43$

AWFW

$$P(\bar{Y}_{40} > 120) = P\left(Z > \frac{120 - 108}{28/\sqrt{40}}\right)$$

*Standardising 120 with 108 and $\sqrt{19.6}$ or
(4.425 to 4.43) or equivalent; allow (108 – 120)*

M1

$$= P(Z > 2.71) = 1 - P(Z < 2.71)$$

Area change; may be implied

m1

$$= 1 - 0.99664 = 0.0033 \text{ to } 0.0034$$

AWFW (0.00336)

(1 – answer) $\Rightarrow$ B1 M1 max

A1

4

[9]

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Q14.

- (a) 12 elephants are a **random sample**
OR are **selected independently**

OE; eg representative

B1

$$\text{Mean } \bar{x} = \frac{39.24}{12} = 3.27$$

CAO

B1

98% $\Rightarrow z = 2.32 \text{ to } 2.33$

AWFW (2.3263)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

Used; must have $\sqrt{n}$ with $n > 1$

M1

Thus $3.27 \pm 2.3263 \times \frac{0.20}{\sqrt{12}}$

ft on $\bar{x}$ and z only

A1ft

Hence 3.27 ± 0.1343

Hence $3.27 \pm (0.13 \text{ to } 0.14)$

OR (3.13 to 3.14, 3.4(0) to 3.41)

AWFW

A1

6

(b) Value of 2.90 is **below / outside** CI

ft on (a); OE

B1ft

Suggests **mean height** of adult male

Asian elephants is less **than / different**

from that / mean height

of adult male **African elephants**

ft on (a); OE

B1ftdep

2

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[8]

Q15.

(a) (i) $P(X < 40) = P\left(Z < \frac{40 - 38}{5}\right) =$

Standardising (39.5, 40 or 40.5) with 38

and ($\sqrt{5}$, 5 or 5^2) and/or (38 - x)

M1

$P(Z < 0.4) = 0.655 \text{ to } 0.66$

AWFW (0.65542)

A1

2

(ii) $P(30 < X < 40) =$
 $P(X < 40) - P(X < 30) =$
 $(i) - P(X < 30) =$
 $(i) - P(Z < -1.6) =$
Difference or equivalent
Standardising other than 40 and 30 $\Rightarrow$
max of M1 m1 A0

M1

$(i) - \{1 - P(Z < +1.6)\} =$
 $0.65542 - \{1 - 0.94520\} =$
Area change

m1

0.6 to 0.601
 AFWW (0.60062)

A1

3

(iii) 75% (0.75) $\Rightarrow z = \pm 0.674$ to ± 0.675
 AFWW (0.6745)

B1

$$z = \frac{x - 38}{5}$$

Standardising x with 38 and 5

M1

$= -0.6745$

Equating z-term to z-value but not using
0.75, 0.25, $|1 - z|$ or $\Phi(0.75) = 0.77337$

m1

$x = 34.6$ to 35
 AFWW

A1

4

(b) (i) 98% $\Rightarrow z = 2.32$ to 2.33
 AFWW (2.3263)

B1

or
 98% $\Rightarrow t = 2.42$ to 2.43
 AFWW (2.426)

(B1)

(Knowledge of the t -distribution is not required in this unit)

$$\text{CI for } \mu \text{ is } \bar{y} \pm (z \text{ or } t) \times \frac{(s_{n-1} \text{ or } s_n)}{\sqrt{n}}$$

Used; must have $\sqrt{n}$ with $n > 1$

M1

Note that $19.1 \times \sqrt{\frac{40}{39}} = 19.34332$

$$19.1 \times \sqrt{\frac{40}{39}} = 19.58974$$

Max of B1 M1 A0ft A1

Thus

$$107 \pm (2.32 \text{ to } 2.43) \times \frac{(19.1 \text{ or } 19.3 \text{ to } 19.4)}{(\sqrt{40} \text{ or } \sqrt{39})}$$

ft on z or t only

A1ft

Hence $107 \pm (7.00 \text{ to } 7.55)$

ie $107 \pm (7 \text{ to } 8)$
or $(99 \text{ to } 100, 114 \text{ to } 115)$

AWFW

A1

4

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(ii) $2\frac{1}{2} \times (\text{mean for adult males eels})$
 $= 2.5 \times 38 = 95$
CAO

B1
↑Dep↑

Since $95 < \text{LCL}$
OE; ft on CI

B1ft
↑Dep↑

Claim appears valid
OE; ft on CI

B1ft

3

[16]

Q16.

$$98\% \Rightarrow z = 2.5758$$

AWFW 2.57 to 2.58

B1

CI width is $2 \times \frac{z\sigma}{\sqrt{n}}$

Used; allow $\frac{z\sigma}{\sqrt{n}}$

M1

$$\text{Thus } 2 \times \frac{2.5758 \times 0.08}{\sqrt{n}} = 0.05$$

OE; ft on z; allow no '2 ×'

A1ft

$$\text{Thus } \sqrt{n} = 8.24256$$

Solving for $\sqrt{n}$ or n

m1

$$\text{Thus } n = 67.9 \Rightarrow 68$$

AWRT; ft on z

A1ft

Thus, to nearest 5, $n = 70$

CAO

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A1

6

[6]

Q17.

(a) (i) Mean, $\bar{x} = 505.2$

CAO; stated or implied

B1

$$99\% \Rightarrow z = 2.57 \text{ to } 2.58$$

AWFW (2.5758)

B1

Or

$$99\% \Rightarrow t = 3.25$$

AWRT (3.250)

B1

(Knowledge of the t -distribution is not required in this unit)

CI for μ is
$$\bar{x} \pm (z \text{ or } t) \times \frac{(\sigma \text{ or } s)}{\sqrt{n}}$$

use of; must have $(\div \sqrt{n})$ with $n > 1$

M1

Thus
$$505.2 \pm 2.5758 \times \frac{6}{\sqrt{10}}$$

ft on $\bar{x}$ and z only

A1ft

or
$$505.2 \pm 3.25 \times \left(\frac{5.96}{\sqrt{10}} \text{ or } \frac{5.65}{\sqrt{9}} \right)$$

ft on $\bar{x}$ only

A1ft

Hence
$$505.2 \pm 4.9$$

use of $t \Rightarrow 505.2 \pm 6.1$

or
 $(500.3, 510.1)$
AWRT

A1

5

- (ii) Given sample:
 3 in 10/ some of packets have weights below 500 grams
or equivalent

B1

Confidence interval:
 $CI > 500$
ft on CI in (a)(i)

B1ft

Conclusion:
 Statement does not appear justified
or equivalent
dependent on both B1 and B1ft

B1 dep

3

- (b) 0.01 or 1%
CAO; or equivalent

B1

1

[9]

Q18.

- (a) $n = 40$ $\bar{x} = 72$ $s = 32$

99% $\Rightarrow z = 2.5758$

AWFW 2.57 to 2.58

B1

CI for μ is $\bar{x} \pm z \times \frac{(s \text{ or } \sigma)}{\sqrt{n \text{ or } (n-1)}}$
Use of
Must have $(\pm \sqrt{n})$ with $n > 1$

M1

Thus $72 \pm 2.5758 \times \frac{32}{\sqrt{40 \text{ or } 39}}$
ft on z only

A1ft

$(58.8 \text{ to } 59.1, 84.9 \text{ to } 85.2)$
AWFW

EXAM PAPERS PRACTICE

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4

- (b) $Y \sim (53, 42^2)$

- (i) Large value of standard deviation,
OE

E1

relative to mean,

suggests negative times are likely
OE

E1

2

- (ii) Due to large sample size

OR

by Central Limit Theorem

$n > 30$
either
CLT

E1

1

(iii) $\bar{Y}$ has mean, $\mu = 53$
CAO

B1

and variance, $\frac{\sigma^2}{n} = \frac{42^2}{60} = 29.4$
CAO; SD = AWFW 5.42 to 5.43

B1

$$P(\bar{Y} < 60) = P\left(Z < \frac{60 - 53}{\sqrt{29.4}}\right)$$

Standardising (AWFW 59 to 60) with 53 and
$$\left(\sqrt{\frac{42^2}{n}} \text{ or } \frac{42^2}{n}; n > 1\right)$$

and/or (53 - x)

M1

= $P(Z < 1.29) = 0.899$ to 0.903
AWFW (0.90165)

A1

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4

[11]

Q19.

(a) Volume $\sim N(\mu, 3.5^2)$

Mean, $\bar{x} = \frac{1830}{12} = 152.5$
CAO ($s_{n-1} = 3.778, s_n = 3.617$)

B1

98% $\Rightarrow z = 2.3263$
AWFW 2.32 to 2.33

B1

CI for μ is $\bar{x} \pm z \times \frac{(\sigma \text{ or } s)}{\sqrt{n}}$
 Use of
 Must have $(\div \sqrt{n})$ with $n > 1$

M1

Thus $152.5 \pm 2.3263 \times \frac{3.5}{\sqrt{12}}$
 ft on $\bar{x}$ and z only

A1ft

(150.1 to 150.2, 154.8 to 154.9)
 AFWW

A1

5

- (b) Evidence, from CI, that mean volume is (above) 150 ml
 ft on CI in part (a); must be clear comparison of mean of 150 with CI

B1ft

In sample, some cans have volumes less than 150 ml
 Or reference to range of can volumes in sample

B1

Thus claim of 150 ml is not justified
 Dependent upon making some comment about mean volume and some comment about individual can volume or range of can volumes

B1dep

3

[8]

Q20.

- (a) (i) Weight, $X \sim N(\mu, 1.6^2)$

95% $\Rightarrow z = 1.96$
 CAO

B1

CI for μ is $\bar{x}(\mu) \pm z \times \frac{\sigma(s)}{\sqrt{n}}$

use of; must have $(\div \sqrt{n})$ with $n > 1$

M1

Thus $49.2 \pm 1.96 \times \frac{1.6}{\sqrt{10}}$
ft on z only

A1ft

(48.2, 50.2)

AWRT; dependent on fully correct expression

A1

4

(ii) CI includes 50 **or** UCL > 50

ft on CI

B1ft

so suspicion is not supported

ft on CI; dependent on first B1

B1ft

2

(b) Require $z \frac{\sigma}{\sqrt{n}} = \text{or } \leq \text{or } < \frac{0.75}{2}$ or 0.75
use of $z \frac{\sigma}{\sqrt{n}}$ n with n unknown

M1

Thus $1.96 \times \frac{1.6}{\sqrt{n}} = \frac{0.75}{2} = 0.375$ or 0.75
ft on z; ignore signs

A1ft

Thus $n = \left(\frac{1.96 \times 1.6}{0.375} \right)^2$
attempt at solving for n

m1

Thus $n = 70$

CAO (18 with no $\div 2$)

A1

4

Note:

Method of 'trial and improvement':

Answer = 18 CAO

B2

Answer = 70 CAO

B4

[10]

Q21.

(a) (i) $X \sim N(\mu_x 3^2)$

$$P(X < 1010) = P\left(Z < \frac{1010 - 1005}{3}\right) =$$

Standardising (1009.5, 1010 or 1010.5) with
($\sqrt{3}$, 3 or 3^2) and/or (1005 – 1010)

M1

$$P(Z < 1.67) =$$

AWRT; ignore sign

A1

$$0.951 \text{ to } 0.953$$

AWFW; (0.95221)

A1

3

(ii) $P(X < 1000) = 1\%$
 $Z_{0.01} = -2.3263$

AWFW 2.32 to 2.33; ignore sign

B1

Also $z = \frac{1000 - \mu_x}{3}$

Standardising 1000 with μ_x and 3 but allow ($\mu_x - 1000$)

M1

Thus $\frac{1000 - \mu_x}{3} = -2.3263$

Equating z-value to z-term; not using 0.01, 0.99 or $|1 - z|$

m1

Thus $\mu_x = 1007$
AWRT

A1

4

(b) $\bar{y} = \frac{16136}{16} = 1008.5$

CAO

B1

95% implies $z = 1.96$

CAO

B1

CI for μ is $\bar{y} \pm z \times \frac{\sigma}{\sqrt{n}}$

Use of; must have $\sqrt{n}$ with $n > 1$
M0 for attempt at using s

M1

Thus $1008.5 \pm 1.96 \times \frac{3}{\sqrt{16}}$
f.t. on $\bar{y}$ and z only

A1f.t.

Thus (1007, 1010)

AWRT; dependent upon fully correct expression for CI

A1dep

5

[12]

Q22.

Weight $X \sim N(\mu, 1.6^2)$

95% $\Rightarrow z = 1.96$

CAO

B1

CI for μ is: $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

use of; must have $\sqrt{n}$ ($n > 1$)

M1

$\therefore 80.8 \pm 1.96 \times \frac{1.6}{\sqrt{64}}$

f.t. on z ; allow 1.6^2 or $\sqrt{1.6}$ but must be $\sqrt{64}$ or 8

A1✓

$\therefore 80.8 \pm 0.392$

$\therefore (80.4, 81.2)$

AWRT

NB attempt at changing σ or σ^2 to unbiased $\Rightarrow$ at most B1
M1

A1

4

[4]

Q23.

(a) Mean, $\bar{x} = 413$
CAO

B1

99% $\Rightarrow z = 2.5758$

AWFW 2.57–2.58

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

use of, must have $\sqrt{n}, n \neq 1$
use of unbiased estimate M1A0A0

M1

$\therefore 413 \pm 2.5758 \times \frac{3}{\sqrt{12}}$

f.t. on $\bar{x}$ and z , allow 3^2 or $\sqrt{3}$ but must be $\sqrt{12}$

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$\therefore 413 \pm 2.2$

$\therefore (410.8, 415.2)$

AWRT Incorrect $z \Rightarrow$ B1BO M1A1f.t.
A0 max

A1

5

(b) As 99% CI excludes 410
f.t. on (a)

B1f.t.

Unlike to be correct

f.t. on (a), dependent on B1

E1f.t.

2

Q24.

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum x = \frac{2848}{16} = 178$$

CAO

B1

CI: $\bar{x} \pm \frac{z\sigma}{\sqrt{n}}$

Use of with $n > 1$

M1

99% $\Rightarrow z = 2.5758$

AWFW 2.57 to 2.58

B1

Thus CI is $178 \pm \frac{2.5758 \times 3}{\sqrt{16}}$

✓ on $\bar{x}$ and z ; not on n

A1f.t.

Thus CI is 178 ± 1.9 or (176,180)

AWRT; dependent upon fully correct expression

A1

EXAM PAPERS PRACTICE [5]

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Q25.

(a) $\bar{x} = 1.33$

1.33 cao

B1

90% confidence interval

$$1.33 \pm 1.6449 \times \frac{0.11}{\sqrt{8}}$$

use of $\frac{0.11}{\sqrt{8}}$

1.6449 (1.64, 1.65)

Completely correct method candidate's

M1 B1 m1

1.33 ± 0.0640

A1

5

$(1.266, 1.394)$

1.33 ± 0.0640 (0.063, 0.065)

or 1.266(1.265, 1.267), allow 1.27

and 1.394(1.393, 1.395), allow 1.39

sc $1.33 \pm 1.895 \frac{0.14}{\sqrt{8}}$

B1M1

$(1.236, 1.424)$

B1m1A0

mixture z & s or t & σ max

B1M1

- (b) Evidence mean content at least 1.20 but
Mean above 1.20

E1

some individual oranges less than 1.20

Some individuals < 1.20

E1

2

(c) $2 \times 1.96 \times \frac{0.11}{\sqrt{72}} \leq 0.03$

Reasonable attempt at equation/inequality

1.96 cao

Completely correct equation/inequality

-allow incorrect z value

M1 B1 m1

$n \geq 206.6$

Method of solution

m1

207 needed

207 cao

A1

5

sc *Trial & Improvement*

205 – 210

B3

200 – 220

B1

[12]

Q26.

(a) $\bar{x} = 135.6$

135.6 cao

B1

95% confidence interval for mean is

$$135.6 \pm 1.96 \times \frac{0.42}{\sqrt{9}}$$

1.96

B1

Use of $\frac{0.42}{\sqrt{9}}$

M1

i.e. 135.6 ± 0.274
135.33 to 135.87

Correct method for interval

m1

Answer given to 4, 5 or 6 sf

A1

135.33 (135.33 to 135.4) and
135.87 (135.8 to 135.9)

Or $135.6 \text{ cao} \pm 0.274$ (0.274 to 0.275)

A1

6

(b) $135.8 \pm 1.96 \times \frac{3.9}{\sqrt{60}}$

Correct method - their z

M1

135.8 ± 0.987
134.81 to 136.79

134.81 (134.8 to 134.820) and

136.79 (136.78 to 136.8)

Or $135.8 \text{ cao} \pm 0.987$ (0.98 to 1)

A1

2

(c) (i) Decrease

B1

1

(ii) Increase - narrower interval → more

E1

likely to identify need for overhaul if $\mu \neq 135.0$.

E1

2

(d) Large variability as in (b) will lead to unsatisfactory production even if mean on target. Might prefer small variability as in (a) even if mean slightly off target.

Large variability unsatisfactory/makes deviation from 135 difficult to detect.

E1

Small variability with mean slightly off target may be preferable

E1

2

[13]

Q27.

(a) (i) 95% confidence interval
1.96 cao

B1M1

$$498.6 \pm 1.96 \times \frac{9.3}{\sqrt{95}}$$

Use of $\frac{9.3}{\sqrt{95}}$

$$498.6 \pm 1.87$$

Correct method - their z

m1

$$496.7 - 500.5$$

*496.7 (496.7 – 496.8) and
500.5 (500.4 – 500.5) or
498.6 ± 1.87 (1.86 – 1.88)
Allow 497 and 500*

A1

4

(ii) 80% confidence interval



$$498.6 \pm 1.2816 \times \frac{9.3}{\sqrt{95}}$$

Method - their z

M1

2

$$498.6 \pm 1.22$$

$$497.4 - 499.8$$

497.4 (497.3 – 497.4) and

499.8 (499.8 – 499.9) or

498.6 $\pm$ 1.22 (1.22 – 1.23)

Allow 497 and 500 but penalise if answers to (i) and (ii) are identical

A1

(b) (i) (A) 0.8

0.8 cao

(B) 0.15

Method

0.15 cao

B1

M1

A1

3

(ii) $0.95 \times 0.2 = 0.19$

Method

0.19 cao

M1

A1

2

[11]

Q28.

(a) $\bar{x} = 48.4$

48.4 cao

B1

99% confidence interval

$$48.4 \pm 2.5758 \times \frac{14.8}{\sqrt{10}}$$

2.5758(2.57 – 2.58)

Use of $\frac{14.8}{\sqrt{10}}$ or $\frac{5}{\sqrt{10}}$

B1 M1

ie 48.4 ± 12.1

m1

36.3 – 60.5

*Completely correct method – candidate's
36.3 (36.3 – 36.4) and 60.5 (60.4 – 60.5)
or 48.4 and 12.1 (12 – 12.1)*

A1

5

(b) (i) 95% confidence interval

$$49.5 \pm 1.96 \times \frac{16.5}{\sqrt{110}}$$

1.96
Method – their z

B1 M1

$$49.5 \pm 3.08$$

49.5 ± 3.08 (3.08 – 3.09)
or 46.42 (46.41 – 46.42)
and 52.58 (52.58 – 52.59)
Allow 46.4 and 52.6

A1

3

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(ii) $49.5 \pm 1.96 \times 16.5$
Method – their z

M1

$$17.2 - 81.8$$

49.5 ± 32.34 (32.3 – 32.4)
or ± 17.2 (17.1 – 17.2)
and 81.8 (81.8 – 81.9)

A1

2

(c) Mean greater than 25 but many
Mean $\neq$ 25

E1

individuals will be less than 25

Mean > 25

E1

Some individuals < 25

E1

3

(d) Interval narrower

Interval narrower

E1

No need for normal assumption

Normal not needed

E1

2

(e) Both samples random so valid suggestion.

Valid

E1

Would be better but not much ($\frac{49.5}{\sqrt{110}}$)

Better

E1

very similar to $\frac{49.5}{\sqrt{120}}$)

Not much

Any three points

E1

3

[18]

Q29.

$$758 \pm 2.5758 \times 6\sqrt{4}$$

2.5758 (2.57 ~ 2.58)

attempt to use $6\sqrt{4}$

completely correct method

B1 M1 M1

$$748 \pm 7.73$$

$$740.3 \sim 755.7$$

$$748 \pm 7.73 (7.72 \sim 7.74) \text{ or}$$

$$740.3 (740 \sim 740.3) \text{ and } 755.7 (755.7 \sim 756)$$

A1

4

[4]

Q30.

(a) $\bar{x} = 627$

cao

B1

$s = 640.38$

640 ~ 641

B1

95% confidence interval for mean is

9 df

B1

$627 \pm 2.262 \times 640.38 / \sqrt{10}$

for 2.262 or 2.26

attempt to use $s\sqrt{10}$

use of t-distribution

completely correct method–

allow incorrect t value

B1 M1 M1 M1

627 ± 458

$169 \sim 1085$

457 ~ 459

or 168 ~ 170 and 1080 ~ 1090. All rights reserved.

A1

8

(b) (i) 3 people drank no water

B1

suggesting a mode of zero which is a long way from the mean / not symmetrical

B1

(ii) amount cannot be negative

B1

3

(c) (i) $960 \pm 1.6449 \times 240 / \sqrt{135}$

for 1.6449 or 1.645 or 1.64 or 1.65 or 1.655 ~ 1.657



for use of $240/\sqrt{135}$

for completely correct method

(allow use of *t*-distribution)

B1 M1 M1

$$960 \pm 34.0$$

$$33.5 \sim 34.5$$

A1

4

$$926 \sim 994$$

$$925.5 \sim 926.5 \text{ and } 993.5 \sim 994.5$$

(ii) first day/night shift unlikely to be typical of all employees

B1

1

[16]

Q31.

(a)

$$\bar{x} = 48.114$$

$$(48.1 - 48.12)$$

B1

95% confidence interval for mean

$$48.114 \pm 1.96 \times \frac{1.8}{\sqrt{7}}$$

$$1.96$$

$$\frac{1.8}{\sqrt{7}}$$

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for use of

completely correct method, 'their' $\bar{x}$, z

B1 M1 m2

$$48.114 \pm 1.333$$

$$46.8 - 49.4$$

$$48.1 (48.1 - 48.12) \text{ and}$$

$$1.33 (1.33 - 1.34)$$

$$\text{or } 46.8 (46.7 - 46.8) \text{ and}$$

$$49.4 (49.4 - 49.5)$$

A1

6

(b) 2.67

$$(2.66 - 2.68) \text{ cao}$$

B1

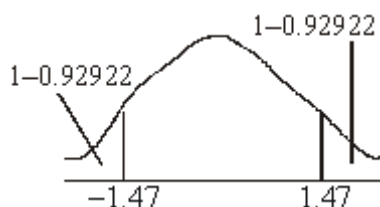
1

(c) $\frac{2z \times 1.8}{\sqrt{7}} = 2$

for any correct equation - allow small slip
e.g. omission of first 2

M1

$z = 1.470$



any completely correct equation
method of solution - requires M mark

m1 m1

% confidence = $100[1 - 2(1 - 0.92922)]$

method for % confidence - requires M mark

m1

= 85.8%

(85 - 86)

A1

5

(d) 0.142

'their' answer to (c)

B1ft

1

[13]

Q32.

(a) $x = 48.11$

$48.1(48 - 48.2)$

B1

95% confidence interval for mean

$48.11 \pm 1.96 \times \frac{1.8}{\sqrt{7}}$

1.96

use of $\frac{1.8}{\sqrt{7}}$

B1 M1

$$48.11 \neq 1.333$$

Wholly correct method

M2

$$46.78 - 49.44$$

$48.11(48.1 - 48.12) \pm 1.33 (1.33 - 1.34)$ or
 $46.8(46.7 - 46.8)$ and $49.4 (49.4 - 49.5)$

A1

6

(b) 2.67

$$2.65 - 2.69$$

B1

1

(c) $2 = 2z \times \frac{1.8}{\sqrt{7}}$

Any reasonable attempt ignore omission of 2
 and other small errors

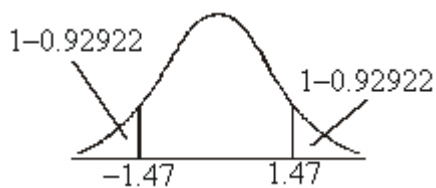
Any completely correct equation involving z

M1 m1

$$z = 1.470$$

method for z requires first M only

©2025 Exam Papers Practice. All rights reserved. m1.



$$\% \text{ confidence} = [1 - 2 (1 - 0.92922)]100$$

Method for probability, their z

M1

$$= 85\%$$

$$85.8 (85 - 86)$$

A1

5

(d) 0.142



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Examiner reports

Q1.

In part (a), the majority of students scored at least three marks. The most common error was to incorrectly calculate the standard error by using 10 as the standard deviation rather than the variance. Some students obtained a probability for the z value or found the z value for a different confidence interval. A small minority of students transposed the digits for the sample mean, obtaining 34.5 instead of 35.4. Some students lost the final mark for not giving their confidence interval to four significant figures.

Most students did not score the mark for part (b). Many understood that Dante would reject the null hypothesis but many did not give a reason or were unclear in their explanation. It was common for students to state that 'it' was not in the confidence interval without explaining what they meant by 'it'. A minority of students wrote that Dante would accept the null hypothesis as the sample mean was in the confidence interval. Some students attempted to calculate the probability for the hypothesis test but none were successful.

Q2.

Full marks were not at all rare here but neither were low or zero marks. In part (a), those students who standardised using $0.15 / \sqrt{12}$ scored at least 2 and often 4 marks. However, those who used 0.15 immediately scored no marks. Some students, presumably as a carry-over from June 2012, attempted, often with little or no success, an

unnecessary change of units. In part (b)(i), a sensible substitution into $\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$ was crucial to scoring marks. Whilst the majority of students did this very well, a significant minority used an incorrect z-value or omitted the divisor of $\sqrt{n}$. Correct deductions, usually following a correct confidence interval, were much in evidence in part (b)(ii). However stating that '4.5 was outside the interval' lost the 2 marks available. (Some students claimed that 4.5 was within (4.57, 4.73)).

Q3.

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This question usually got students off to a good start with most finding the confidence interval correctly in part (a). Occasional errors were seen with the use of an incorrect z-value or the omission of $\sqrt{25}$. In part (b), despite similar requests appearing on many previous papers, students often did not make an explicit comparison of the stated 20 kg claim with their confidence interval. The use of "it" or "mean" was not accepted nor, of course, was a comparison of 20 kg with 19.9 kg.

Q4.

Most students found the confidence interval correctly. Occasional errors, in addition to simple numerical ones, were the use of $\sum x$ rather than $\bar{x}$, an incorrect z-value or the omission of $\sqrt{25}$. Despite similar requests appearing on many previous papers, students often did not make an explicit comparison of the stated 20 kg claim with their confidence interval. The use of "it" or "mean" was not accepted nor, of course, was a comparison of 20 kg with $\bar{x}$. Again, despite similar questions appearing on previous papers, many students remained unable to distinguish between the two structures in part (b). Thus

standardising using $\frac{0.35}{\sqrt{10}}$ in both parts was common as was the failure to raise a

probability to the power 10 in part (b)(ii). Nevertheless, a significant proportion of students, not all of whom were necessarily high achievers, scored full marks in part (b).

Q5.

Most candidates found all but part (b)(ii) of this question very challenging. In part (a), the common incorrect answers were verbose about university salaries or simply that 24.0 was too large when compared to 45.8. Even those candidates who calculated, for example $(45.8 - 2 \times 24.0)$, simply stated that a normal distribution, instead of salaries, could not take negative values. Answers to part (b)(i) were slightly better with candidates indicating a 'large sample' so 'Central Limit Theorem applied'. Some candidates lost both marks for 'if sample is large' whilst others lost a mark for suggesting that other than the sample mean could be assumed to be normally distributed. There were many correct answers to part (b)(ii) though some candidates lost marks for multiplying 24.0 by some combination of n and $(n - 1)$ or for an incorrect z -value. In answering part (c), a majority of candidates attempted the comparison of 55 with their CI or UCL. However, far too many then did the

same with 60 rather than compare $\frac{6}{50} = 12\%$ with 25%. This perhaps suggested that they had not read the question with sufficient care.

Q6.

The better students generally scored full marks in part (a) usually by working in litres. However, a large proportion of students provided answers that scored at most 2 marks. Errors seen were the use of 5 instead of 5.05, the use of 75 or 0.75 instead of 0.075, an incorrect z -value or the omission of the divisor of $\sqrt{36}$. The one mark for answering the first claim in part (b) was only available to those better students who had obtained a correct confidence interval in part (a). Even so, some such students made no comparison using the stated mean of 5 litres but merely commented that the sample mean of 5.05 was greater than 5. This was very disappointing particularly given similar requests on previous papers. In answering the second claim in part (b), some students again referred to their

confidence interval and so scored no marks. Of those that correctly evaluated $\frac{8}{36}$ as 22%, many were not sufficiently explicit in comparing this with 10% and so scored only 1 of the 2 marks available.

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Q7.

In part (a), it was expected that candidates would understand that for a sample of any size from a normal population with known variance, the reference distribution is $N(0, 1)$. Far too many candidates based their choice of distribution on the small value of n , and consequently incorrectly used t_{n-1} as their reference distribution. This resulted in a loss of all marks for this part of the question.

In part (b)(i), most candidates correctly used a t -distribution. Some candidates still used the z -distribution, with $z_{\text{crit}} = 6449$ often seen.

Although most candidates correctly found $\bar{x} = 550$, there were far too many who thought

that $s\sqrt{\frac{334}{9}}$ or, even worse, $s = \sqrt{334}$.

In part (b)(ii), far too many candidates failed to state the level of significance, even though this was specifically asked for in the question. Others incorrectly thought that the level of significance was 90%. Many candidates did not justify their comments by correctly stating

that 545 was not in the confidence interval; incorrect statements, such as 'it is not in the CI', gained no credit.

Q8.

In part (a)(i), it was expected that candidates would use 'area of rectangle = 1'. This was not always the case, with integration methods again being employed where they are totally unnecessary and time consuming. Those who did manage to write down $\text{area} = 10u \times 0.011 =$ sadly then often seemed unable to complete the required algebraic manipulation in order to obtain the given answer.

In part (a)(ii), most candidates managed to correctly find $E(X)$ and $\text{Var}(X)$ by using the suggested formulae. Some however did not leave their answers in terms of as requested and consequently lost marks.

Part (a)(iii) was very poorly answered except by the most able candidates. Far too many candidates either did not know the formula $C = \pi D$, or failed to realise that this formula needed

to be used, and so simply attempted to find $E\left(X + \frac{10}{X}\right)$ and $\text{Var}\left(X + \frac{10}{X}\right)$, often incorrectly. However, part (b) was usually very well done.

The vast majority of candidates realised that, as the variance of the given distribution was known, a z -value (not a t -value) had to be used.

Q9.

This was another very well answered question, with the majority of candidates scoring full marks. One or two candidates struggled unsuccessfully to solve their correct equations, in terms of $\sqrt{n}$, for n , whilst a few others left their answers as 96 or indicated that, to the nearest 10, this was 90 instead of 100.

Q10.

Overall this was the worst answered question on the paper with most answers revealing a marked lack of understanding of much of the material examined. As a result, very few candidates scored more than about 5 marks. Answers to part (a) were generally very disappointing, with most candidates at best only able to score 1 mark for finding $\bar{t}$ as an unbiased estimate of μ . The fact that the 50 times needed to be a random sample

appeared unknown to most candidates, as did the fact that $\frac{\sum (t - \bar{t})^2}{49}$ was an unbiased estimate of σ^2 ; knowledge of the latter is a key area for future improvement. Failure to determine, or even attempt to determine, an unbiased estimate of σ^2 inevitably had a knock-on effect in part (b), as almost all candidates, whilst aware of the formula required, obtained an inaccurate or, more often, a totally incorrect confidence interval. In part (c), most candidates realised that they needed to compare 60 minutes with their intervals, but many then failed to translate their comparisons into a correct statement about Wyn's belief. Answers to part (d) revealed that an understanding of the correct interpretation of a confidence interval was not well understood. Common incorrect answers were 0, 1, or 0.99.

Q11.

Most candidates scored the 5 marks available in parts (a) and (b) but then scored no marks in part (c). When errors occurred in part (b), it was usually for the use of $\sum x$ rather than $\bar{x}$, an incorrect z-value or the lack of the divisor $\sqrt{n}$. It should be noted that as the value of σ was given, then the use of $\sqrt{n} - 1$ was incorrect, as were attempts at changing 8 to an unbiased estimate.

In answering part (c), most candidates simply restated their answers to part (b) rather than using $n - 1$.

Q12.

Candidates were very well-prepared for part (a) and, as a result, most of them scored all 5 marks. Of the small minority who lost marks, the usual error was the use of an incorrect z-value. Here, the rounding of 1003.99 to 1003 or that of 1056.01 to 1057 were treated as irrelevant subsequent work and so were not penalised.

Many candidates scored the 2 marks in part (b) for a correctly reasoned comment on the claim. However, weaker candidates based their comments on a comparison of 1030 with their confidence intervals or suggested that the shovel always collected more than 1000 kg.

Q13.

Most candidates knew how to find a confidence interval for μ and, despite somewhat unusual notation at times, many were able to score full marks. Where marks were lost, it was usually for using an incorrect z-value. Answers to part (b) revealed an improvement in candidates' knowledge of the distribution of the sample mean, in particular that

$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ rather than σ^2 . Such candidates usually scored full marks, but a minority lost a mark for rounding the z-value to 2.7 rather than 2.71.

Q14.

Most candidates simply ignored the request for a necessary assumption in part (a) or stated that 'Heights' or 'Elephants', or 'The sample' or even 'They' were normally distributed. An answer needed to indicate that the 12 elephants were 'A random sample' or were 'Selected independently' or were 'Representative'. Construction of the confidence interval was invariably correct. However a small minority of candidates lost marks for using an incorrect formula, a calculated value of s^2 rather than 0.20 or even an incorrect value for the sample mean.

In part (b), many candidates lost marks because their answers did not justify their (often too definitive) statements. A clear comparison of 2.90 with a calculated confidence interval was required and conclusions had to refer to mean heights, not heights in general.

Q15.

It was most gratifying to note the many candidates scoring 14 or more marks (out of 16) on this question. In part (a)(i), 'corrections to 40' were thankfully very rare with the result that most candidates scored full marks. Again in part (a)(ii), full marks was the norm although a small minority of candidates failed to allow for the negative sign of 1.6 and so evaluated $0.94520 - 0.65542$ (loss of all 3 marks). The most common mistake in part

(a)(iii), seen even on the best scripts, was to equate $\frac{x - 38}{5}$ to + 0.6745 instead of -0.6745 (loss of 1 mark).

Despite comments on previous Examiner Reports, there still remain candidates who are apparently unaware of Table 4.

There were many correct answers given for the confidence interval in part (b)(i). The most common error was the use of an incorrect z-value, often 2.0537 (Table 4 yet again).

Thankfully very few candidates omitted to divide the standard deviation by $\sqrt{40}$. In part (b)(i), candidates had to first identify that $2.5 \times 38 = 95$ before any marks were awarded.

Most candidates did indeed do this but a few evaluated $\frac{107}{2.5}$. Having obtained the value of 95 candidates then had to compare it against their confidence interval, not 107, in order to gain marks for a (follow-through) correct conclusion. Thus it was pleasing to see the many instances of candidates scoring all 3 marks.

Q16.

Centres are to be congratulated on their candidates' many fully correct answers to this question. Where marks were lost, it was usually for omitting the factor 2 in the formula for the confidence interval width or, less frequently, for deducing that 67.9 was 75 to the nearest 5.

Q17.

In part (a)(i), many candidates scored all 5 marks. When this was not the case, it was usually due to an incorrect z-value or the omission of $\sqrt{10}$. In part (a)(ii), the awarding of full marks was rare. Many candidates did not make it clear that they were comparing 500 (and not the sample mean) with their confidence interval, failed to indicate that 3 in 10 packets were underweight or stated no overall conclusion. Correct answers to part (b) were also rare. The word 'Hence' together with an allocation of 1 mark should have suggested that lengthy working, as produced by many candidates, was not required.

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Q18.

Many candidates scored full marks in part (a). A small minority of candidates used an incorrect z-value or omitted the divisor $\sqrt{n}$. In part (b)(i), it was rare to see an answer scoring both marks. Whilst there was invariably some reference to the size of the standard deviation, very few candidates linked this to the size of the mean and hence to likely negative values. Most candidates scored the mark in part (b)(ii) for phrases such as 'n large' or 'Central Limit Theorem'. However, in part (b)(iii), many candidates scored few, if

any, marks through standardising 60 using 53 and 42 instead of 53 and $\frac{42}{\sqrt{60}}$. Perhaps candidates need to be made more aware of the distribution of the sample mean?

Q19.

Many candidates appeared ill-prepared for this type of routine question. As a result, in part (a) the success rate on the straightforward confidence interval calculation was disappointing. Most candidates found the correct value for the sample mean but some candidates used a z-value of 2.0537, rather than 2.3263, whilst others omitted the divisor

$\sqrt{12}$ or used the sample standard deviation instead of the given value of 3.5. However, in general, well-prepared candidates scored all 5 marks.

In answering part (b), most candidates paid no heed to the fact that 3 marks were available and so opted for general subjective assumptions that usually scored no marks. Thus few candidates made use of their calculated confidence interval and even less gave a correct interpretation of it. There was also minimal reference to the sample data. As a result, even the best candidates, who recognised the relevance of both the sample data and their confidence interval, were reluctant to commit to a conclusion.

Q20.

Only the strongest candidates scored full marks on this question. In part (a)(i) common

errors were use of an incorrect z-value or, more critically, use of σ instead of $\frac{\sigma}{\sqrt{n}}$.

Somewhat surprisingly, correct answers to part (a)(ii) were quite rare. Many candidates failed to comment on the fact that the quoted 50 kg was within their (correct) confidence interval. All too often reference was made to the fact that the majority of their confidence interval was below 50 kg, or that their confidence interval included the mean (presumably 49.2 kg?) or that 50 kg was above 49.2 kg. Answers to part (b) often included a correct

method even if lacking in final accuracy. The usual error was to equate $z = \frac{\sigma}{\sqrt{n}}$ to 0.75 rather than to 0.375, usually resulting in a loss of 1 mark. Of concern was the number of candidates who followed $\sqrt{n} = a$ by $n = \sqrt{a}$. Infrequent attempts at trial and improvement brought success to a small number of candidates but, needless to say, an absence of any associated working resulted in a score of 4 or 0.

Q21.

This question proved to be a good source of marks for the majority of candidates, where full marks was not uncommon.

In part (a)(i), apart from standardising 1009.5, using $\sqrt{3}$ or 32 for σ , or rounding $\frac{5}{3}$ to 1.66 rather than 1.67, most answers were fully correct. In part (a)(ii), there was a large proportion of correct answers. Where errors occurred, it was usually through using $z = +2.3263$ or standardising 1010 or 1. Again, in part (b), full marks were the norm. Here,

mistakes included equating $\bar{x}$ to 16136, z to 1.6449 or σ to $\sqrt{\frac{16 \times 9}{15}}$.

Q22.

Many candidates scored full marks on this first question. Rare errors included using 2 or 1.645 rather than 1.96, attempting to correct 1.6 for bias, and omitting $\sqrt{64}$ as a divisor. Individually, such errors resulted in a loss of at least 2 of the 4 marks available.

Q23.

This question elicited good answers from most candidates and there were few common errors. In part (a), some candidates used $z = 3$ whilst others calculated an unbiased estimate of variance from the twelve values or tried to adjust the given population standard deviation for bias. Most candidates who found a confidence interval in part (a)

were then able to make a sensible (follow-through) attempt at part (b), though a small minority compared 410 with 413 or with the twelve individual values.

Q24.

Many candidates scored full marks in this question. A few candidates used 2.3263 for 99%, whilst a similar small number attempted to find an unbiased estimate of σ^2 by

calculating $\frac{3^2 \times 16}{15}$ or even $\frac{3 \times 16}{15}$.

Q25.

Part (a) caused few problems. Candidates who used the t-distribution correctly were penalised one mark on the grounds that they had not used all the information in the question. However, candidates who mixed t and σ or z and s were treated less sympathetically. Part (c) was less straightforward but had a high success rate. More candidates than expected used part (b) to demonstrate that they did not understand the nature of confidence intervals. A common misconception was that a 90% confidence interval would contain 90% of the population.

Q26.

Parts (a) and (b) were well answered. Unlike previous years, there was a mark in part (a) for giving the answer to an appropriate degree of accuracy (4, 5 or 6 significant figures were accepted in this case). Those candidates who hedged their bets by giving alternative answers to different levels of accuracy did not receive this mark unless all their answers were to 4, 5 or 6 significant figures.

Most candidates made reasonable attempts at parts (c) and (d), although answers to part (d) revealed that many candidates did not realise that confidence intervals apply to means and are not intervals within which a given percentage of the population lies.

Q27.

The calculations posed few problems. Although the rubric states that answers should 'normally be given to three significant figures', candidates should realise that it is not normal for an 80% and a 95% confidence interval to be identical. There were many good answers to part (b), although a common error was to answer (b)(i)(B) as if the samples were independent, as they were in (b)(ii).

Q28.

Parts (a) and (b)(i) were well answered. Only a minority of candidates were able to answer part (b)(ii) correctly, many merely repeated their answer to part (b)(i). Others, apparently influenced by a question on a previous paper, calculated the width of the confidence interval.

The interpretation in part (c) was generally poor. Many candidates failed to make any connection between the confidence interval and the mean, believing rather that 95% of scores lie in the 95% confidence interval. The answers to parts (d) and (e) were better. In part (e) there was some concern as to the validity of combining two samples. However as both samples were stated to be random samples the combined sample would also be a random sample.

Q29.

Most accepted this as an easy starter and scored full marks. The most common error was to use incorrect z -values. A small number used t -values, which is incorrect since the standard deviation has clearly not been calculated from the sample.

Q30.

Most candidates were well prepared for part (a) although a few failed to use the t -distribution. Part (b) was not generally well answered, particularly part (b)(ii) where only a small minority pointed out that it is not possible to consume negative amounts of water and so the normal distribution cannot provide an exact mode.

All parts of (c) were well answered. It was expected that, as the sample was large, candidates would use z -values and the given standard deviation. Some used t -values.

Some interpreted the standard deviation as $\sqrt{\frac{\sum (x - \bar{x})^2}{n}}$ and applied a correction to obtain their value.

Most candidates made good attempts at parts (c)(i) and (c)(ii) although quite a few stated that because the sample was large the distribution (rather than the distribution of the sample mean) would be approximately normal.

Q31.

Most candidates correctly used the given standard deviation. A few used the t -distribution. This, while not incorrect, is less appropriate as it fails to make use of all the available information. In this case, it also made the remainder of the question more difficult.

In part (b), many candidates failed to realise that in order to obtain an answer correct to 3 significant figures it is necessary to use confidence limits to more than 3 significant figures.

Most candidates correctly found the z -values in part (c) - some by trial and error. However considerably fewer candidates were able to complete the calculation of the percentage confidence. Despite this, most candidates picked up the mark in part (d) for an answer which was correct given their answer to part (c).

Q32.

In part (b), many candidates failed to realise that in order to obtain an answer correct to 3 significant figures it is necessary to use confidence limits to more than 3 significant figures.

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